CS328 A2 — Activity Classification: Exploration Report

Data: 615 Class_Data recordings (Sensor Logger phones, gravity removed) and 35 Sample_Data recordings (wrist device, 5 people). Every number comes from the submitted notebook (*results.json*). "Split" is the notebook's 70/30 random window split. Because it puts windows of the same recording on both sides, we also report **grouped CV**: 5-fold CV on Class_Data with whole recordings held out, which is what a new user would see. We chose features and models on grouped CV, not on the leaky split.

§1 Feature exploration

Window length (baseline features, depth-5 entropy tree): 2 s 52.7%, 3 s 53.8%, 5 s 51.1%, 10 s 53.9% on the split, and 48-50% grouped CV at every length, so window length is not the bottleneck. We chose **5 s, non-overlapping**: about 9 steps per window, 612 of 615 recordings keep at least one window (10 s keeps 603), and a typical 40 s recording gets 5-10 votes.

Two rounds of features. Round 1 used only the magnitude (rhythm, spectrum, noise): grouped CV 48.9% → 57.1%. Round 2 went back to the three axes and added a *vertical proxy* (the motion projected on its main axis, close to vertical while walking whatever the phone's rotation) and features of what Sensor Logger's imperfect gravity removal leaves behind (residual mean and slow <1 Hz drift per axis): → **64.7%**. Each row below adds one feature to the 7 baseline statistics. "Change, split" = depth-5 tree on the notebook split; "Change, grouped CV" = random forest (100 trees) with recordings held out; "Drop from final" = grouped-CV change when the feature is removed from the final 20 (64.7%). F1 values are the depth-5 tree's.

Feature	Class it helps discriminate	Change, split	Change, grouped CV	Drop from final	Rationale
min	static vs gait	—	—	+0.1	(baseline) Deepest dip of the detrended magnitude: large on every step, ≈0 at rest. Top importance (0.11).
vert_period	upstairs (F1 0→0.28), walk	+3.1	+4.3	-0.9	Lag of the strongest autocorrelation peak of the vertical proxy: a full stride, slower on stairs.
vert_std	jogging (0.79→0.87), downstairs	+2.4	+1.9	-0.4	Spread of vertical motion: bounce height. Rotation invariant, unlike std_z.
slow_fast_ratio	standing (0.48→0.52), downstairs	+3.4	+3.1	-0.3	Slow (<1 Hz) drift vs fast (>5 Hz) jitter. Standing sways; a still phone only has sensor noise.
vert_stride_reg	lying (0.09→0.21), downstairs	+1.4	+2.3	-0.5	Autocorrelation at the second peak: how alike consecutive strides are (Moe-Nilssen regularity).
slow_z	sitting (0.65→0.70), standing	+2.2	+2.5	-0.4	Slow drift on the z axis: gravity-removal residual, so it depends on how the phone is tilted.
var_frac_y	upstairs (0→0.19)	+1.7	+2.7	-0.5	Share of the motion on the y axis: where the motion goes relative to the phone.
kurt	none alone	+0.7	+0.2	+0.7	Spiky (impacts, taps) vs smooth magnitude.
mean_z	lying (0.09→0.19), downstairs	+2.9	+0.2	-0.3	Residual gravity left on z after Sensor Logger subtracts its gravity estimate: a weak tilt cue.
acf_peak	downstairs (0→0.53)	+3.3	+1.9	-1.0	Regularity of the magnitude: periodic gait vs irregular motion.
mean_y	lying (0.09→0.21)	+2.4	+1.7	-1.5	Residual gravity on y. Largest single drop when removed from the final set.
var_frac_z	lying (0.09→0.21)	-0.7	+1.6	-0.4	Share of the motion on z.
vert_kurt	upstairs (0→0.34)	+0.9	+1.6	+0.1	Peakedness of vertical acceleration: sharp landings vs smooth push-off.
logvar	none alone	+0.0	-0.4	-0.5	Log of the magnitude variance: intensity on a log scale (rest spans decades). Second-highest importance.
pca2	lying (0.09→0.19)	-0.3	+0.5	-0.2	Share of motion on the 2nd principal axis: flat vs line-like motion.
vert_skew	downstairs (0→0.42)	+1.6	+1.2	+0.0	Asymmetry of vertical acceleration: push-off up vs landing down.
slow_frac_y	none alone	+0.0	+1.1	+0.1	Share of slow drift on y; helps only together with slow_z.
slow_frac_z	none alone	+0.1	+0.1	-0.5	Share of slow drift on z.
med	none alone	—	—	+0.2	(baseline) Median of the detrended magnitude.
corr_yz	downstairs (0→0.42)	+1.9	+0.1	-0.1	Correlation of the y and z axes: coordinated vs independent motion.
step_period (dropped)		+2.4	+3.5	—	Round-1 magnitude version of vert_period; superseded.
spec_entropy (dropped)		+3.1	+3.5	—	Round 1, +3.5 alone, but redundant with acf_peak + vert_* once those are in.
hf_frac (dropped)		+1.9	+1.5	—	Round 1; covered by slow_fast_ratio.
std_z (dropped)		+2.4	+0.5	—	Depends on placement; replaced by variance fractions.
dom_freq (dropped)		+3.0	+2.3	—	Harmonics make it jump between step and stride rate.
rms (dropped)		-0.1	-0.4	—	Duplicates std and min.
round-1 8 (magnitude only)		+3.9	+8.2		grouped CV 57.1%
compact 8 = first 8 above		+5.5	+11.2		grouped CV 60.0%
compact 12		+6.7	+15.0		grouped CV 63.9%

Feature	Class it helps discriminate	Change, split	Change, grouped CV	Drop from final	Rationale
final 20		+6.2	+15.8		grouped CV 64.7%
baseline + all 37		+6.1	+14.3		grouped CV 63.2%

How we chose. Greedy forward selection over 90 research candidates with recordings held out (research/exp15.py), then re-implemented per window in add_my_features (37 candidates kept there); the order above is the order of selection. The rubric suggests 5-8 features: the first 8 alone reach 60.0%, and we kept 20 (64.7%) because the remaining errors are in the hardest classes and every point there counts. **Honest note:** leave-one-out changes grouped CV by ≤1.6 points for every feature (several within ±0.5, i.e. fold noise), so the information is spread across correlated features rather than carried by any one of them. Adding all 37 candidates is *worse* than our 20 (noise features dilute the forest).

§2 Model exploration

Same 20 features and 5 s windows for every model. Tree and Extra Trees hyperparameters come from the notebook grid searches (chosen on the Class_Data split); the others are standard settings.

Model	Key hyperparameter(s)	Split	Grouped CV	Class→Sample	Sample→Class	F1 up / lying	Notes
Decision Tree	max_depth=5, entropy	57.3%	52.4%	61.3%	47.0%	0.07 / 0.21	Baseline, interpretable; almost never predicts upstairs.
Decision Tree	max_depth=15, entropy	64.2%	50.2%	56.6%	39.3%	0.33 / 0.31	Grid winner on the split, worse on grouped CV and transfer: memorises recordings.
Random Forest	n_estimators=300	74.9%	64.7%	67.1%	48.4%	0.44 / 0.34	Close second; best Class→Sample and Sample→Class.
Extra Trees	n_estimators=200, max_features=0.5, max_depth=20	78.0%	65.7%	66.8%	47.7%	0.46 / 0.32	**Chosen.** Best split and grouped CV; random thresholds average out person-to-person noise.
HistGradientBoosting	lr=0.05, 300 iterations	76.5%	63.4%	64.3%	48.2%	0.47 / 0.37	Best upstairs/lying F1, but overfits the leaky split more than it gains.
Gradient Boosting	n_estimators=100, max_depth=3	70.8%	62.1%	65.4%	43.4%	0.41 / 0.36	Slower, weaker version of the above.
kNN	k=15, StandardScaler	64.0%	56.4%	53.2%	33.7%	0.37 / 0.24	Distances across 20 mixed-scale features are noisy; worst transfer.
Logistic Regression	C=1, StandardScaler	60.7%	58.3%	62.2%	39.8%	0.29 / 0.21	Linear boundaries: fine for intensity, not for static classes.
SVM	RBF, C=3, StandardScaler	69.6%	62.0%	62.0%	32.1%	0.43 / 0.24	Good in-domain, worst Sample→Class.

Which won and why. The tree ensembles (Extra Trees 65.7%, random forest 64.7%, boosting 62-63%) and the RBF SVM (62%) form a top group about 13 points above the single tree (52.4%). We picked **Extra Trees**: best on the notebook split (78.0%) and on grouped CV, and within a point of the best on Class→Sample. With 20 features a single tree has to spend each split on one threshold, so classes that need several weak cues together (upstairs: slower vertical period plus kurtosis plus variance share; lying: residual gravity on two axes) get few leaves; hundreds of randomised trees cover those regions. **The gain was not equal across classes.** Class→Sample, Extra Trees raised upstairs F1 0.31→0.51, downstairs 0.45→0.62, walk 0.51→0.71, sitting 0.51→0.68, standing 0.60→0.74; jogging stayed at 0.98, and lying got worse (0.24→0.03): the ensemble votes a still wrist into "sitting", the more common still class in the training data. The tuned-tree row shows the cost of tuning on the leaky split: the deeper tree wins the split (+6.9) but loses on grouped CV (–2.2) and on Sample_Data (–6.1).

§3 Universal pipeline

Step	Choice
Preprocessing	Sampling rate per file = 1 / median timestamp step (50-101 Hz in the data). Magnitude of (x,y,z) detrended by the recording mean (calc_magnitude); 2nd-order Butterworth low-pass at 5 Hz, zero-phase (remove_noise).
Windows	5 s, non-overlapping, clock-aligned via DataFrame.resample; windows with <80% of the expected samples are dropped.
Features (20)	min, vert_period, vert_std, slow_fast_ratio, vert_stride_reg, slow_z, var_frac_y, kurt, mean_z, acf_peak, mean_y, var_frac_z, vert_kurt, logvar, pca2, vert_skew, slow_frac_y, slow_frac_z, med, corr_yz (definitions in add_my_features)
Classifier	ExtraTreesClassifier(n_estimators=200, max_features=0.5, min_samples_leaf=1, max_depth=20, random_state=42), trained on Class_Data
Aggregation	Majority vote of the window labels per recording (recording = activity folder + file name, because Sample_Data reuses 026.csv…030.csv in every folder); ties broken by the summed predict_proba.
Performance	Notebook split **78.0%**. Grouped CV on Class_Data: **65.7%** of windows, **66.8%** of recordings. Class→Sample: 69.1% of windows, **74.3%** of recordings.

How we decided. One axis at a time. Window length first, with baseline features, because the grouped CV showed it barely mattered. Features next, judged with a random forest under grouped CV: the 30% random split leaks windows of the same recording into test and kept preferring deep trees that then did worse on new recordings and on Sample_Data. Round 1 (magnitude only) plateaued at 57%; the error analysis said the confusions were posture and stairs, which live in axis information, so round 2 added the vertical proxy and gravity-residual features. Model last: Extra Trees was best or tied-best on every protocol except Sample→Class, where nothing reaches 49%. For aggregation, majority vote beat confidence weighting (74.3% vs 71.4% Class→Sample).

§4 Confusion matrix + per-class analysis

Final pipeline, Class_Data, grouped 5-fold CV, window level (5370 windows, accuracy 65.7%). Rows = predicted, columns = actual, as in the template.

Predicted ↓ / Actual →	walk	upstairs	downstairs	jogging	standing	sitting	lying
walk	**803**	191	119	15	11	12	0
upstairs	64	**219**	109	1	4	1	0
downstairs	75	116	**326**	12	4	0	1
jogging	3	7	2	**339**	0	2	0
standing	12	21	6	0	**498**	128	111
sitting	4	1	1	0	198	**1169**	410
lying	0	0	2	0	43	156	**174**
recall	84%	39%	58%	92%	66%	80%	25%

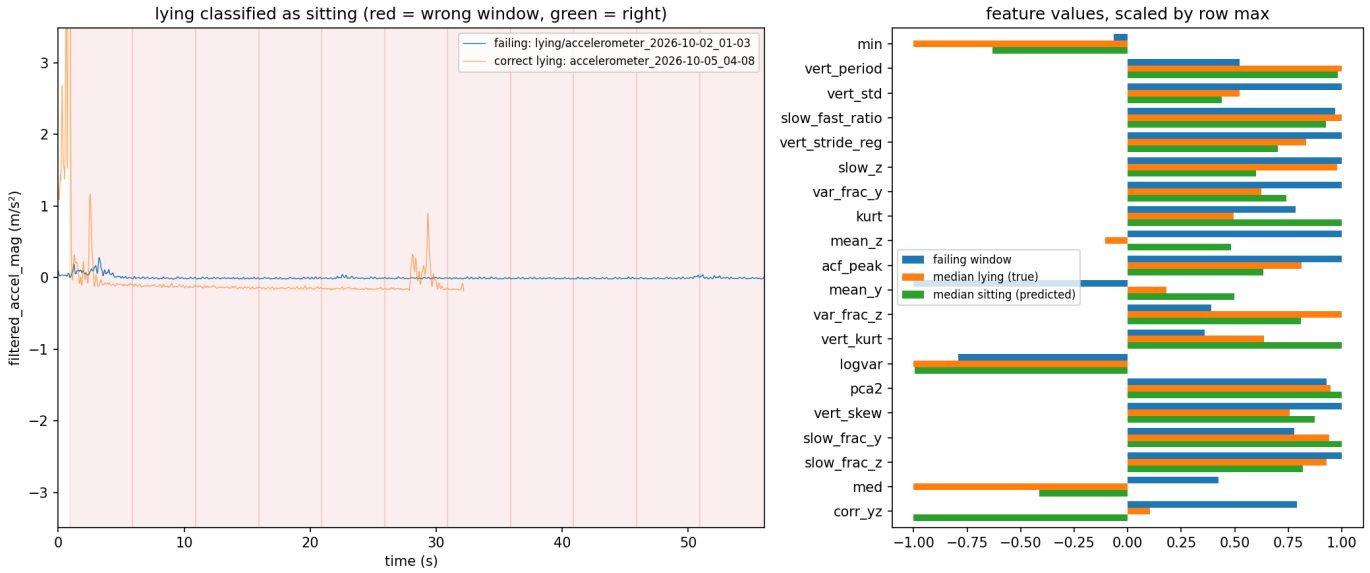
Recording level (majority vote, same CV): 66.8%; per-class recall walk 94%, upstairs 46%, downstairs 66%, jogging 92%, standing 71%, sitting 78%, lying 20%.

Cell (actual → predicted)	Count	Cause	Axis
lying → sitting	410	A still phone reads ≈0 whatever the posture. 43 of 82 lying and 68 of 116 sitting recordings have <0.01 m/s² of slow motion, i.e. the phone is resting (research/exp14.py). Medians: vert_std 0.033 vs 0.028, logvar -7.16 vs -7.13.	Feature
standing → sitting	198	Standing usually sways (median logvar -4.5 vs -7.1), but a phone put down on a table while the person stands is indistinguishable from sitting.	Feature
upstairs → walk	191	Same intensity and cadence: min -2.37 vs -2.38, vert_std 2.54 vs 2.62, vert_period 1.12 vs 1.14 s.	Feature
sitting → lying	156	The mirror of lying→sitting; the forest splits the "phone at rest" recordings between the two classes on weak residual-gravity cues.	Feature
sitting → standing	128	Fidgeting, or holding the phone while sitting, adds sway-like slow motion.	Feature
downstairs → walk / upstairs	119 / 109	A slow, careful descent loses the sharp landings (vert_kurt, vert_skew) that set downstairs apart; the magnitude has no sign of up vs down.	Feature
upstairs → downstairs	116	Both stairs are slower and more irregular than walking; telling up from down needs the direction of vertical motion, which is lost with gravity.	Feature
lying → standing	111	Recordings with the phone held in the hand while lying: hand tremor looks like standing sway.	Feature
walk → down / up	75 / 64	Heavy or brisk walkers produce stair-like impacts. The single tree confuses far more of these than the forest (walk F1 0.60 vs 0.76), so for the tree this part is model side.	Feature (model for DT)

Overall: the remaining error is on the **feature side**. The five best model families top out at 62-66% grouped CV, and the confused pairs have nearly equal medians. The clear model-side effect is the single tree vs the ensembles (§2): about 13 points, mostly on walk and the stairs.

§5 Top failure modes + edge cases

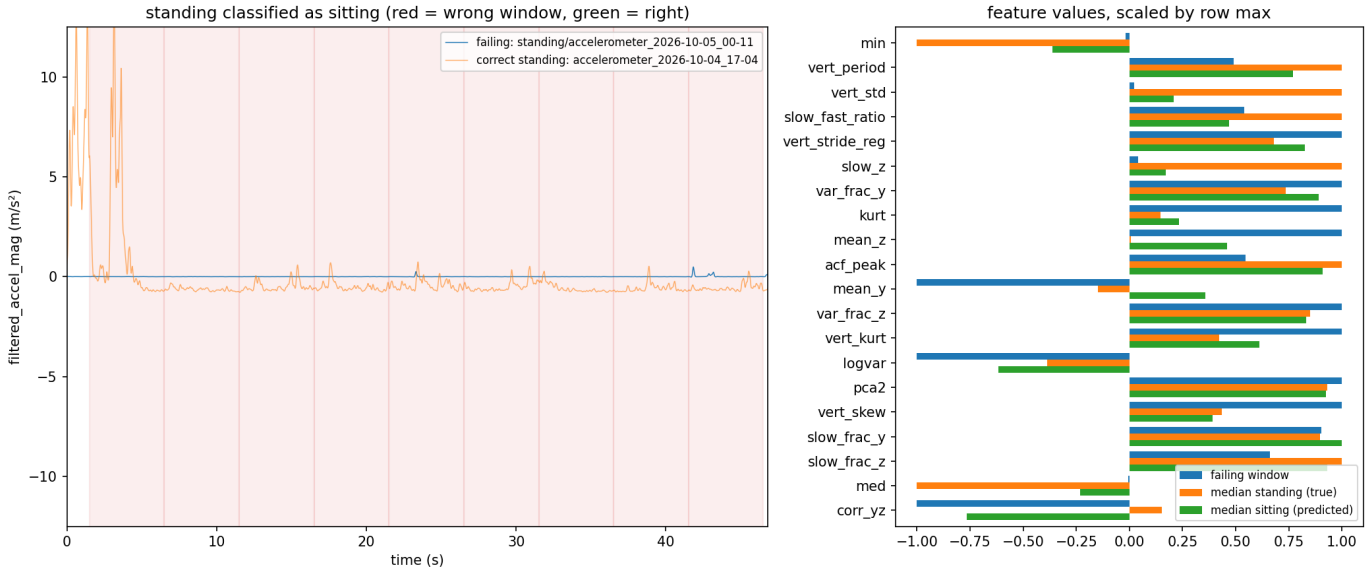
Failure 1: lying classified as sitting (410 windows, 59% of all lying windows; only 25% of lying is right). Example: *lying/accelerometer_2026-10-02_01-03-19.csv*, 11/11 windows → sitting.



Left: the failing trace (blue) is flat: the phone is resting. A correctly classified lying recording (orange) for comparison; red bands = windows predicted wrong. Right: the failing window's features vs the class medians, each row scaled by its max. vert_std 0.063 (lying 0.033, sitting 0.028), slow_z 0.008 (0.008 / 0.005): the failing window is just as close to sitting as to lying.

Mechanism: Sensor Logger subtracts gravity, so the one thing that differs between lying and sitting, which way the body (and phone) points, is not in the signal. When the phone rests on a bed or table, all that is left is sensor noise. **Feature problem:** the two classes have the same feature distribution. **Fix:** also log Sensor Logger's gravity sensor (or the raw accelerometer) and add the tilt of the gravity vector in phone coordinates.

Failure 2: standing classified as sitting (198 windows, 26% of standing windows). Example: *standing/accelerometer_2026-10-05_00-11-28.csv*, 9/9 windows → sitting.



The failing recording (blue) is almost perfectly flat: logvar -11.6 vs a standing median of -4.5 and a sitting median of -7.1; vert_std 0.003 vs 0.134 / 0.028. A person standing with the phone in a pocket sways (orange); this phone was not on a body.

Mechanism: the phone was set down (or held perfectly still) while the participant stood, so the recording contains no body sway at all, which is the only cue for standing in gravity-free data. Of all still recordings, sitting is the most common label, so the model calls it sitting. **Feature problem** (arguably a labelling/protocol problem). **Fix:** require a fixed placement (front trouser pocket) in the study protocol, or add the gravity-vector tilt, which separates upright from seated thighs.

Third largest, for completeness: upstairs → walk (191 windows, 34%). Climbing at a natural pace has the same cadence and magnitude as walking; the difference (lifting the body each step) is on the vertical axis, and the vertical proxy only recovers part of it. Fix: barometer (a flight ≈ 0.3 hPa) or projecting on the gravity vector.

Edge cases in the data: recordings from 4.5 s to 27 min (the 27-min sitting file alone gives 323 windows); a few phones at 50-58 Hz, hence the per-file sampling-rate estimate; one lying file with a 20 s logging gap (those windows are skipped); a "walk" recording named "activity_1_-_cooking"; 3 recordings shorter than 5 s produce no window.

Why not 80%? We tried: 90 candidate features, 9 model families, 2-10 s windows, trimming recording ends, per-recording weighting and recording-level context features (research/exp6, exp9-18). The ceiling on unseen recordings is about 66-67%, and our numbers match published subject-independent results: UniMiB-SHAR reports 73% (9 classes) and 57% (17 classes) with leave-subject-out vs much higher with random splits. ExtraSensory reports leave-user-out per-class scores from 0.63 (sitting) to 0.77 (walking). On the leaky notebook split the same pipeline scores 78%; that gap is the leakage, not skill.

References: Bao & Intille 2004 (window features + decision trees for accelerometer HAR); Kwapisz, Weiss & Moore 2011, WISDM (phone in pocket; stairs hardest); Anguita et al. 2013, UCI-HAR (time/frequency features, gravity-separated signals); Moe-Nilssen & Helbostad 2004 (gait regularity from trunk-acceleration autocorrelation); Micucci et al. 2017, UniMiB-SHAR (arxiv.org/abs/1611.07688); Vaizman et al. 2018, ExtraSensory (arxiv.org/abs/1609.06354).